The Daily Brief

Django News Application - Capstone Project Completion Report

Project	Django News Application
Database	MariaDB migration completed
API	Token-authenticated REST API with role-based access
Verification	28 automated tests passed

Project overview

The Daily Brief is a Django news application for readers, independent journalists, and editors. Readers can view approved articles, subscribe to publishers and journalists, and access subscription-based content. Journalists submit articles and create newsletters. Editors review and approve submissions.

Core implementation completed

- Created a custom user model with Reader, Journalist, and Editor roles.
- Added Publishers, Articles, Newsletters, subscription relationships, and approval logs.
- Added Django groups that stay aligned with each user's role, including the required permissions for articles and newsletters.
- Implemented article approval, subscriber email notifications, and internal approval logging.
- Connected browser pages for public articles, newsletters, journalist submission, login, and editor review.

API and access control

The application includes a token-authenticated REST API. The following endpoints support the required workflow:

Endpoint	Purpose
POST /api/token/	Obtain a token using a username and password.
GET, POST /api/articles/	List approved articles; journalists submit articles.
GET /api/articles/subscribed/	Reader-only access to subscribed content.
GET, PUT, PATCH, DELETE /api/articles/<id>/	Retrieve or manage an article according to role.
POST /api/articles/<id>/approve/	Editor approval of an article.
GET, POST /api/newsletters/	List or create newsletters.
GET, PUT, PATCH, DELETE /api/newsletters/<id>/	Retrieve or manage a newsletter.

Security and data work

- Secured the internal approval callback with an environment-based token in the X-Callback-Token header.
- Used constant-time token comparison for callback validation and rejected missing or incorrect tokens.
- Stored MariaDB credentials and callback values in a local .env file excluded from Git.
- Exported the original SQLite data, migrated the MariaDB schema, loaded the fixture, and confirmed the Django system check passed.
- Added a .gitignore, .env.example, requirements.txt, and README for a cleaner GitHub submission.

Testing and manual demonstration

Automated tests were written with Django REST Framework's testing tools. The final test run completed successfully with 28 tests passing and no Django system check issues.

Area checked	Examples covered
Authentication	Valid token login, rejected incorrect password, protected profile.
Role permissions	Reader denied article creation; journalist submission; editor management and approval.
Subscriptions	Reader receives only approved articles from subscribed publishers and journalists.
Newsletters	Only approved articles accepted; author editing and deletion rules checked.
Callback workflow	Missing and invalid tokens rejected; valid secured callback creates an approval log.
Website workflow	Public page filtering, journalist web submission, and editor web approval.

Postman evidence

An importable Postman collection and safe environment template are included in the postman folder. They demonstrate public endpoints, role-specific token login, reader restrictions, journalist submission, editor approval, and newsletter creation. The collection contains response tests for each expected outcome.

Submission checklist

- Submit the contents of the Capstone -News_app folder from Documents/Codex.
- Include the Python source code, templates, migrations, README, Postman folder, and this report.
- Do not submit the myenv folder, local .env file, SQLite database, backup fixture, passwords, or generated tokens.
- Use Postman screenshots only after hiding passwords and tokens.

Prepared as supporting documentation for the Django News Application capstone submission.